PlatePrep

User manual — from field photos to annotation-ready images

Bioē Project — Institute of Biology, University of Campinas (Unicamp), Dept. of Biochemistry and Molecular Biology.
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What it is

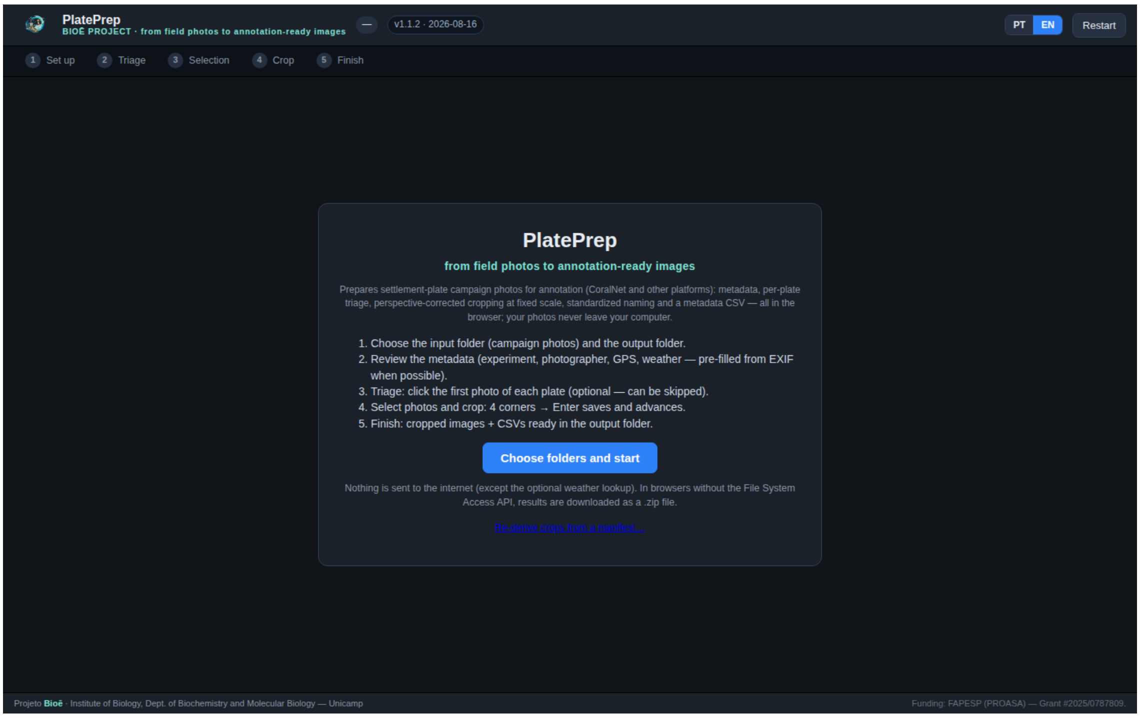
PlatePrep prepares **settlement-plate** campaign photos for annotation in CoralNet and other platforms, in a single 5-step browser workflow: campaign metadata, per-plate triage, selection, perspective-corrected cropping at fixed scale, and generation of named images + CSVs in the output folder. Interface in Portuguese and English (PT/EN toggle in the header). Photos never leave your computer; the only external call is the optional weather lookup (Open-Meteo).

Requirements

- **Google Chrome** (desktop). Direct folder access uses the *File System Access* API; in browsers without it, results are downloaded as a .zip at the end.
- Photos must be **available on the computer** (in Google Drive, mark the folder as *Available offline*).
- The weather lookup needs internet at click time; all fields can be filled in manually.

Getting started

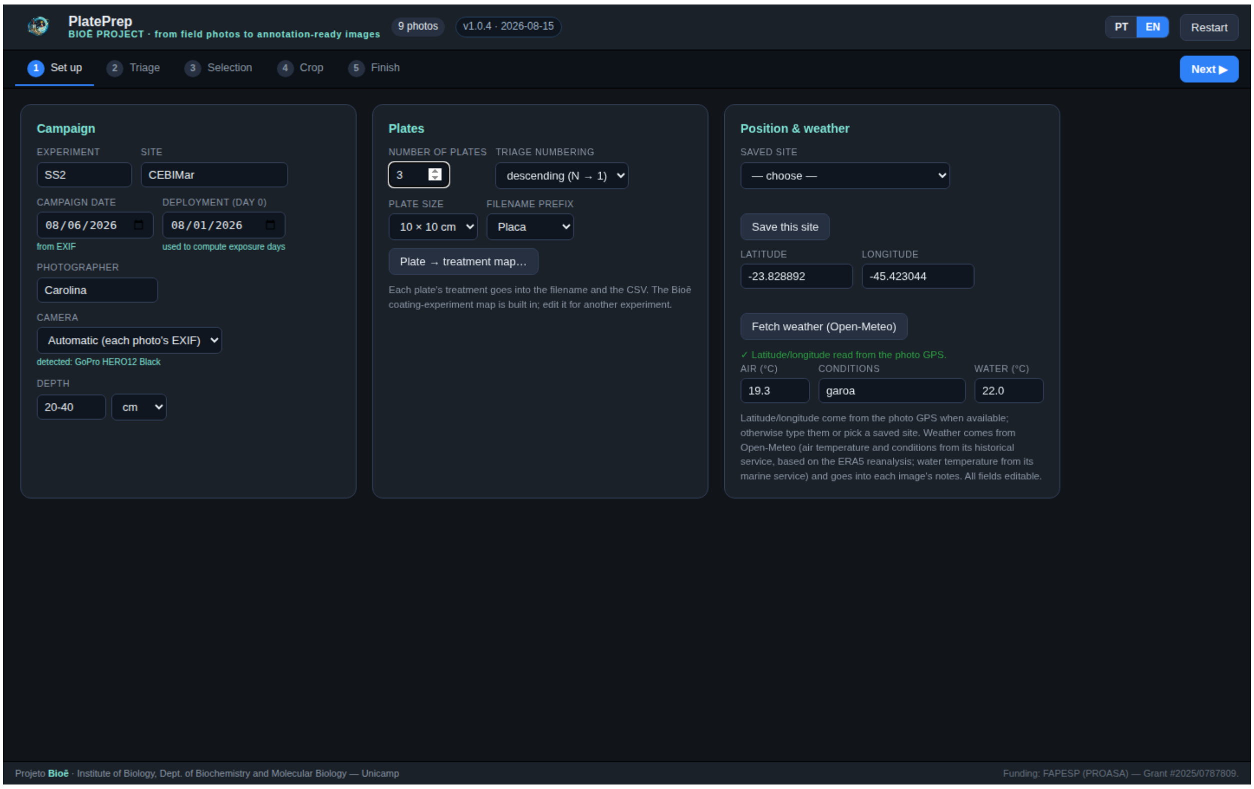
Open PlatePrep in Chrome (the project server address, or the `index.html` file). On the start screen, click **"Choose folders and start"** and select the **input folder** (the campaign photos, JPG/PNG) and the **output folder**. The program reads the EXIF of all photos and moves to step 1.



Start screen, with the PT/EN language toggle in the header.

Step 1 — Set up

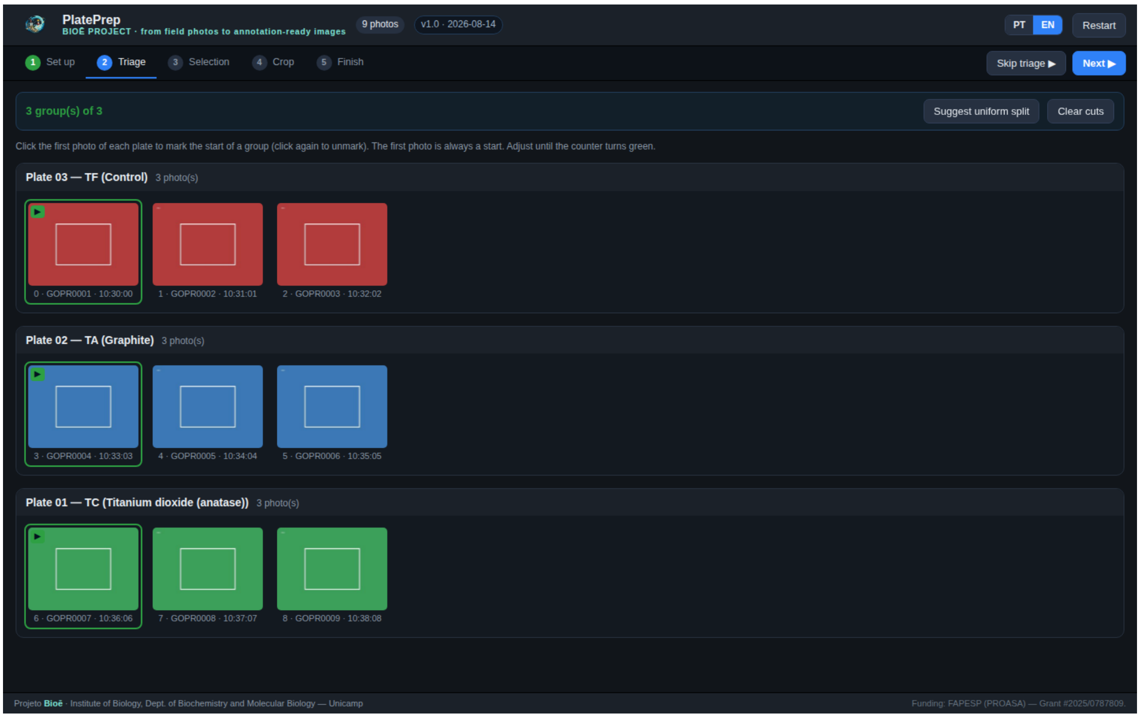
The campaign metadata, typed once, composes each crop's file name and the metadata CSV. **Campaign date**, **camera** and **latitude/longitude** arrive pre-filled from the photos' EXIF/GPS ("from EXIF" / "from photo GPS" tags); if the camera has no GPS, type lat/lon. The **deployment date (day 0)** is used to compute exposure days. Under **Plates** you set the number of plates, triage numbering direction, plate size (fixes the crop scale) and the **plate → treatment map** — the Bioē experiment map is built in and editable for other experiments. Under **Position & weather**, when the photos carry no GPS you pick a **saved site** (the program stores named sites in the browser) or type the coordinates; one click fetches the day's weather from **Open-Meteo** (air temperature and conditions from its historical service, based on the ERA5 reanalysis; water temperature from its marine service). The three fields are editable and go into each image's notes.



Step 1: campaign, plates, position & weather — EXIF-derived fields are tagged.

Step 2 — Triage (optional)

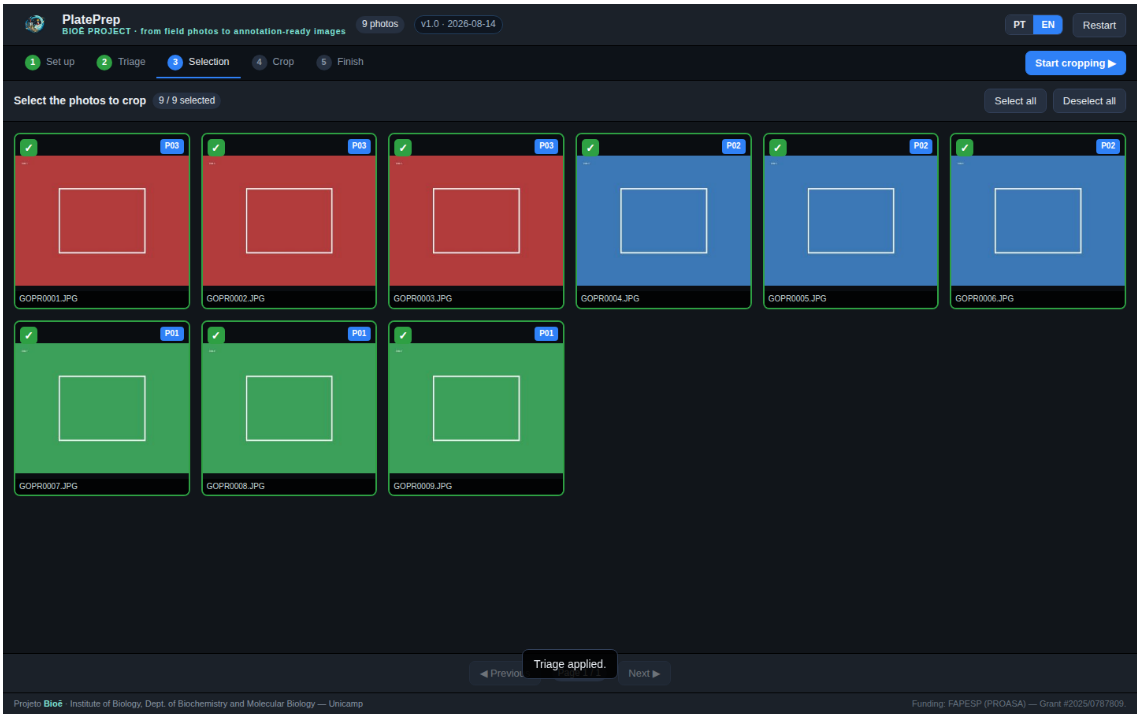
Photos appear in sequence with their EXIF time. Click the **first photo of each plate** to mark a group start (click again to unmark); the program groups the rest and numbers the plates (descending by default). The counter turns **green** when the number of groups matches the number of plates. **"Suggest uniform split"** gives a starting point. Triage feeds the crop step — each photo's plate arrives pre-filled — and produces `triage_mapping.csv`. You can also **skip** this step (button in the header) and pick plates manually while cropping.



Step 2: groups per plate; each group shows plate, treatment and photo count.

Step 3 — Selection

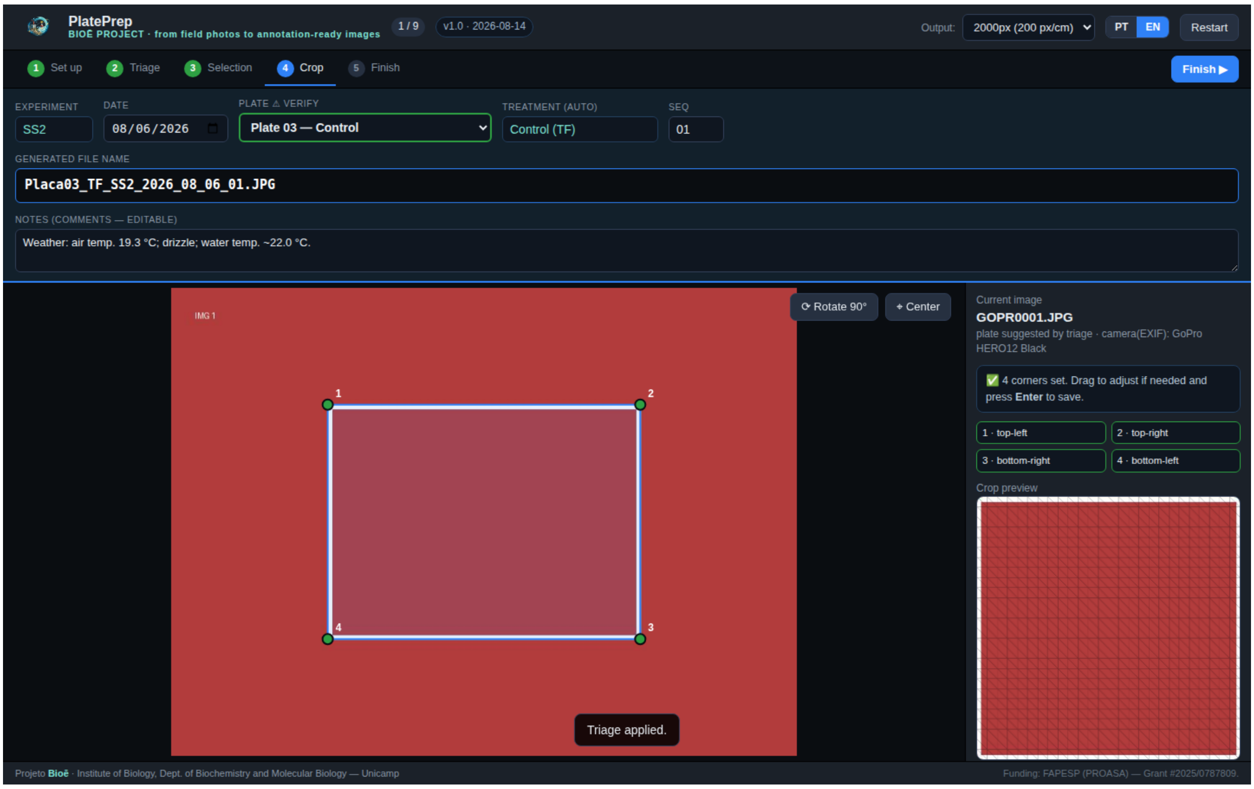
A gallery of all photos; the blue tag shows the plate assigned in triage. Check the photos to be cropped — photos already cropped (present in the output folder) start unchecked.



Step 3: photo selection, with each thumbnail's triage plate.

Step 4 — Crop

For each photo, click the plate's **4 corners** (clockwise from top-left). The preview shows the rectified crop; the **loupe** helps with precision, and points can be dragged or nudged with the arrow keys. In the top bar, the **plate comes pre-filled from triage** (verify!); treatment, file name and sequence are automatic, and the notes already carry the weather. **Enter** saves and advances. Cropping uses **fixed-scale homography**: the whole image corresponds to the plate side (10, 20 or another value in cm), which enables measurements. The photo is read at **native resolution** and every crop pixel is computed through the exact homography (no mesh).

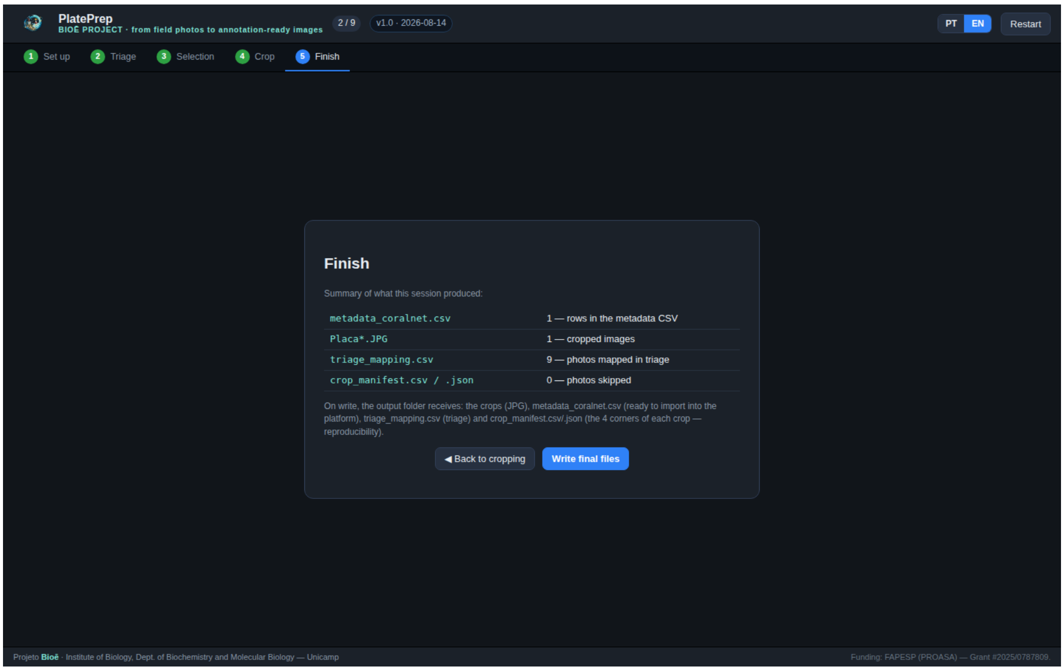


Step 4: 4 corners set, preview on the right, automatic name and notes in the top bar.

Action	How
Save & next	Enter
Undo point / clear	Backspace / Esc
Skip photo	s
Zoom / scroll	Ctrl+wheel / wheel (Shift = horizontal)
Pan the image	Alt + drag, or scrollbars
Center / rotate 90°	C / R
Fine-tune a point	arrows (Shift = 5 px)

Step 5 — Finish

The summary shows what was produced; **"Write final files"** writes everything to the output folder (or downloads a .zip in fallback mode).



Step 5: summary and final file writing.

What is generated

File	Contents
`Plate##_T?_EXP_YYYY_MM_DD_##.JPG`	Square crops, perspective-corrected, fixed scale. The physical scale (px/cm) is written into the JPG's JFIF header — nothing is drawn on the image.
`metadata_coralnet.csv`	One record per image, with the columns CoralNet recognizes (Name, Date, Experiment, Site, Treatment, Exposure_Days, Plate, Height (cm), Latitude, Longitude, Depth, Camera, Photographer, …, Comments).
`triage_mapping.csv`	Original photo → plate/treatment/sequence (triage record).
`crop_manifest.csv / .json`	The 4 corners of every crop, source-file and working-canvas dimensions, plate size and px/cm — re-derive any crop and measure between-operator variability.
`PlatePrep_ImageJ_scale.ijm`	ImageJ/Fiji macro that sets the global scale of the crops (Set Scale: px = plate cm). Run once per session (Plugins > Macros > Run…).

Tips

- **"Photo won't load":** check that the file is available offline (Google Drive).
- **No GPS on the camera:** type latitude/longitude in step 1 — weather and the CSV use those values.
- **Recent date:** Open-Meteo historical data (ERA5 reanalysis) can lag a few days; if the lookup comes back empty, fill the weather in manually.
- **Wrong orientation:** use Rotate 90° (R) before marking corners.
- **Crop already exists:** the program asks whether to replace, create new or discard (shortcuts 1/2/3).
- **Measuring in ImageJ/Fiji:** ImageJ's JPEG reader ignores JFIF density; run the PlatePrep_ImageJ_scale.ijm macro from the output folder (or Analyze > Set Scale: distance = image side in px, known = plate side in cm, Global). Photoshop, GIMP, QGIS and Python/PIL read the scale straight from the JPG.
- **Re-deriving crops:** on the welcome screen, "Re-derive crops from a manifest…" reproduces earlier crops from the source photographs and the manifest (corners rescaled to the current resolution), also writing a log and a rendering audit.

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